- *Continuous deployment and hosting:* Amplify provides seamless integration with AWS services like AWS CodeCommit, AWS CodeBuild, and AWS CodePipeline for automating the deployment process. Developers can easily configure continuous deployment pipelines and host their applications on AWS services like Amazon S3, Amazon CloudFront, or AWS Amplify Console.

AWS generative AI – Bedrock

Amazon Bedrock is an innovative service designed to facilitate the development and scalability of generative AI applications. These applications have the remarkable ability to generate text, images, audio, and synthetic data in response to prompts. With Amazon Bedrock, customers gain convenient access to foundation models (FMs) — the powerful, ultra-large machine learning models that are at the core of generative AI. These foundation models are sourced from top AI start-up providers such as AI21, Anthropic, and Stability AI, along with exclusive access to AWS's own Titan family of foundation models.

We will gather a comprehensive data set of airport-related images, including runways, gate side indoor launches, signages, immigration counters, carousels and other relevant visual elements, ensuring the data set covers a wide range of perspectives, lighting conditions, and variations to train a robust generative AI model.

We will utilise generative AI models such as a generative adversarial network (GAN) or variational autoencoder (VAE) to train on our airport data set. The model learns the visual patterns, structures, and characteristics of airports from the collected images. We train the model to generate realistic and diverse airport scenes and objects dynamically based on the boarding pass after the passenger has checked in. Once the passenger clicks the navigate button on our app, the AR experience module gets loaded into the app dynamically. The personalised AR scenario may differ for each passenger. The produced model will be realistic with visually coherent objects placed within the AR environment.

Since most of the airlines and airports already use AWS and store data, by using the Bedrock generative AI we will be able to get more training data. Also, AWS Bedrock can run in a small set of training data and is serverless. Accessing the AWS Bedrock can happen via an API call.

Viro React: An open source AR

Viro React is a platform that enables developers to create immersive augmented reality (AR) and virtual reality (VR) experiences using React Native. It provides a set of tools, components, and APIs that simplify the development process and allow developers to build cross-platform AR/VR applications for iOS, Android, and virtual reality headsets.

Key features of Viro React: Viro React's powerful components simplify AR/VR development across platforms. Here are some of its key features and benefits.

- *Cross-platform compatibility:* Viro React supports building AR/VR applications for both iOS and Android platforms, allowing developers to create consistent experiences across devices. It also offers compatibility with popular virtual reality headsets like Google Cardboard, Daydream, etc.
- *React Native integration*: Viro React leverages the React Native framework for building mobile applications, which enables developers to leverage their existing React Native skills and codebase. This allows for faster development and easier code sharing between mobile and AR/VR projects.
- *Powerful AR and VR components:* It provides a wide range of ready-to-use AR and VR components that simplify the creation of interactive and immersive experiences. These components include 3D objects, 360-degree media, particle systems, physics simulations, audio playback, and more.
- *Scene and object management:* It can easily manage scenes, objects, and their interactions within the AR/VR environment. The platform offers a scene graph-based architecture, allowing for hierarchical organisation of objects and intuitive manipulation of their properties.
- *Physics and animation:* Viro React incorporates physics-based simulations and animation capabilities, enabling realistic interactions and dynamic movements within the AR/VR environment. Developers can apply physics forces, collisions, and constraints to objects, as well as define complex animations and transitions.
- *AR Core and ARKit support:* It seamlessly integrates with AR Core (for Android) and ARKit (for iOS), providing access to advanced AR features such as plane detection, motion tracking, light estimation, and face tracking. Developers can leverage these capabilities to create engaging AR experiences.
- *Performance optimisation:* The platform offers performance optimisations to ensure smooth and responsive AR/VR experiences. It leverages native rendering capabilities, hardware acceleration, and dynamic occlusion culling techniques to deliver high-performance graphics and interactions.
- *Interaction and input:* It supports various input methods, including touch gestures, device sensors (such as gyroscope and accelerometer), voice commands, and controller input (for VR headsets). Developers can implement interactive features